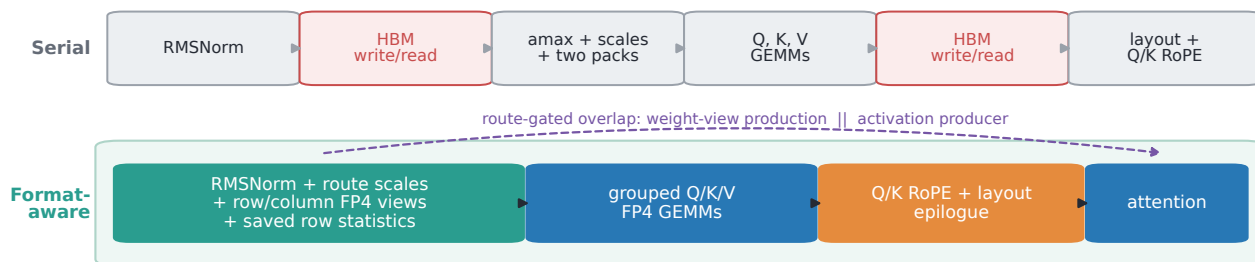
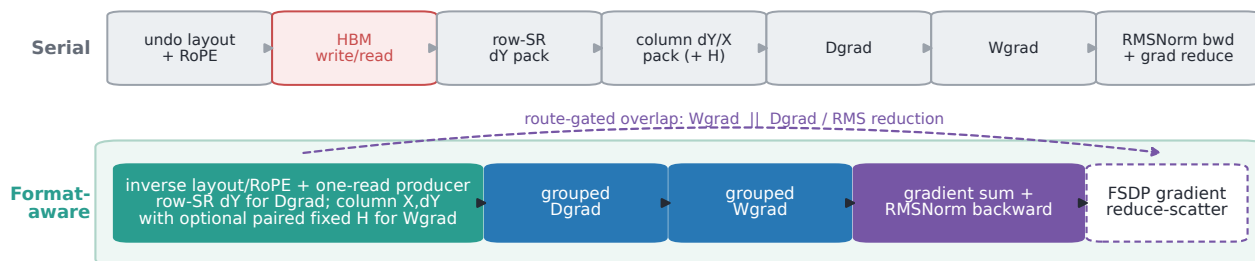


# Fusing the repeated producer--consumer boundaries of a Llama block

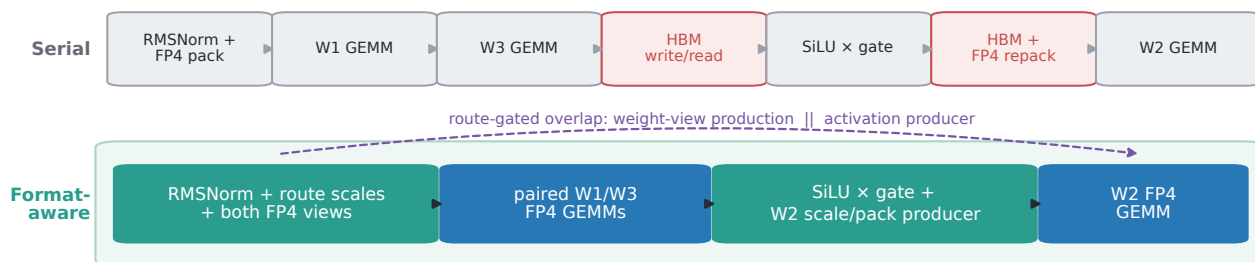
## A QKV forward



## B QKV backward



## C FFN forward



## D FFN backward

